Voltage Controlled LFO

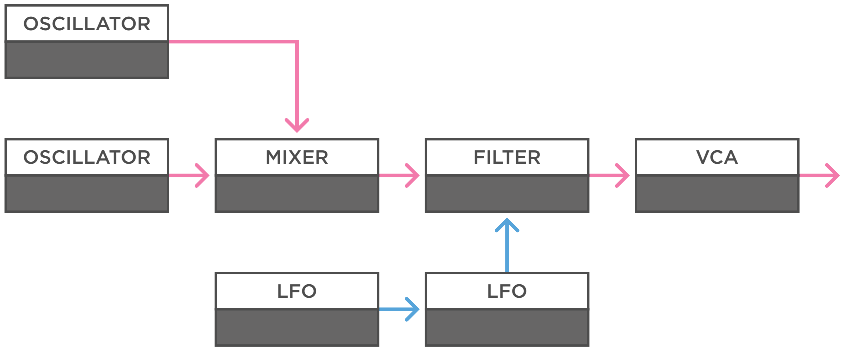
Dynamic Modulation

Audio

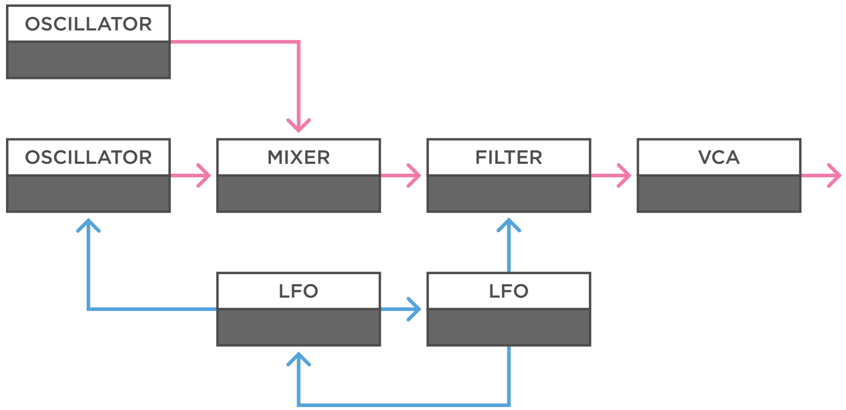
Control Voltage

Trigger / Gate

MONOTRAIL
TECH TALK



(a)



(b)

a - Let's start with a simple drone patch. Here we use two oscillators, a mixer to mix them together, and a single filter to shape the drone. At the end there's a VCA to control the volume, but it's opened all the time to create a constant drone.

If we add two LFOs we can send the output of one of them to the filter, but instead of having that LFO run steady, we introduce some modulation from a second LFO to create more dynamic movement.

b - Because there's two of them, we can exploit that to create even more complex sounds. Of course, you can send a LFO to any parameter you want. Within a drone you could try things like wave-folders, delay time and reverb amount for example. But let's have the second LFO influence the pitch of one of the oscillators. Just a little bit, to create some noticeable oscillator drifting. And if we want, we can even introduce some cross modulation between the two LFOs.

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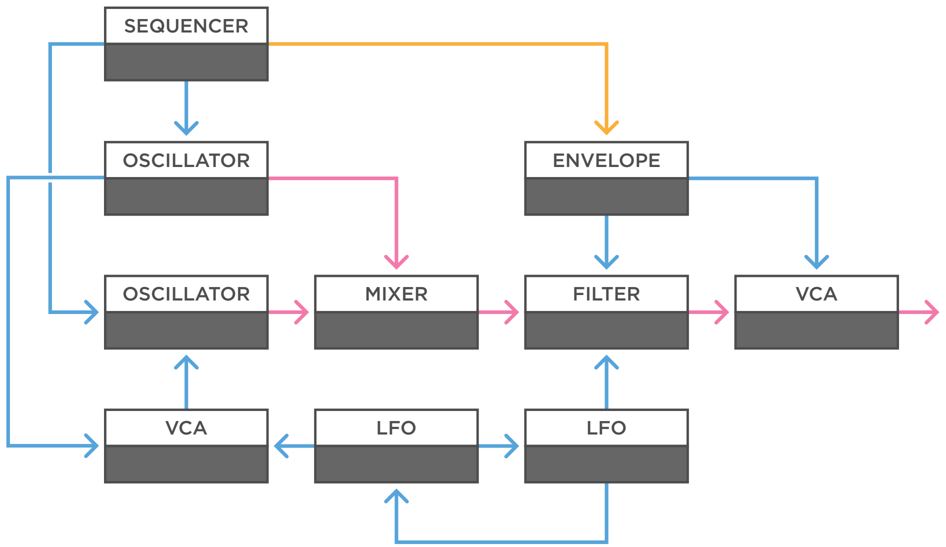
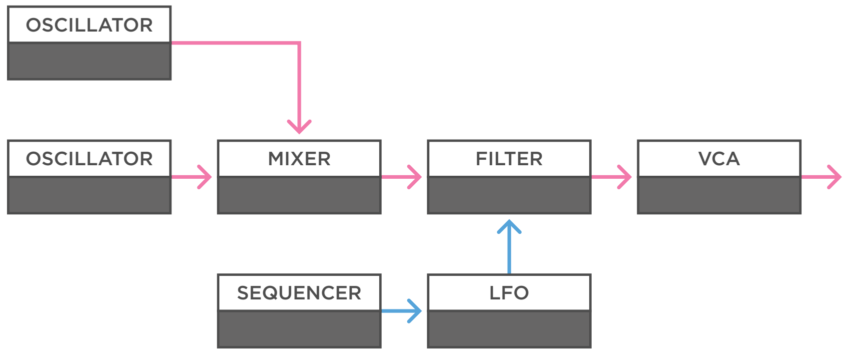


(a)



(b)

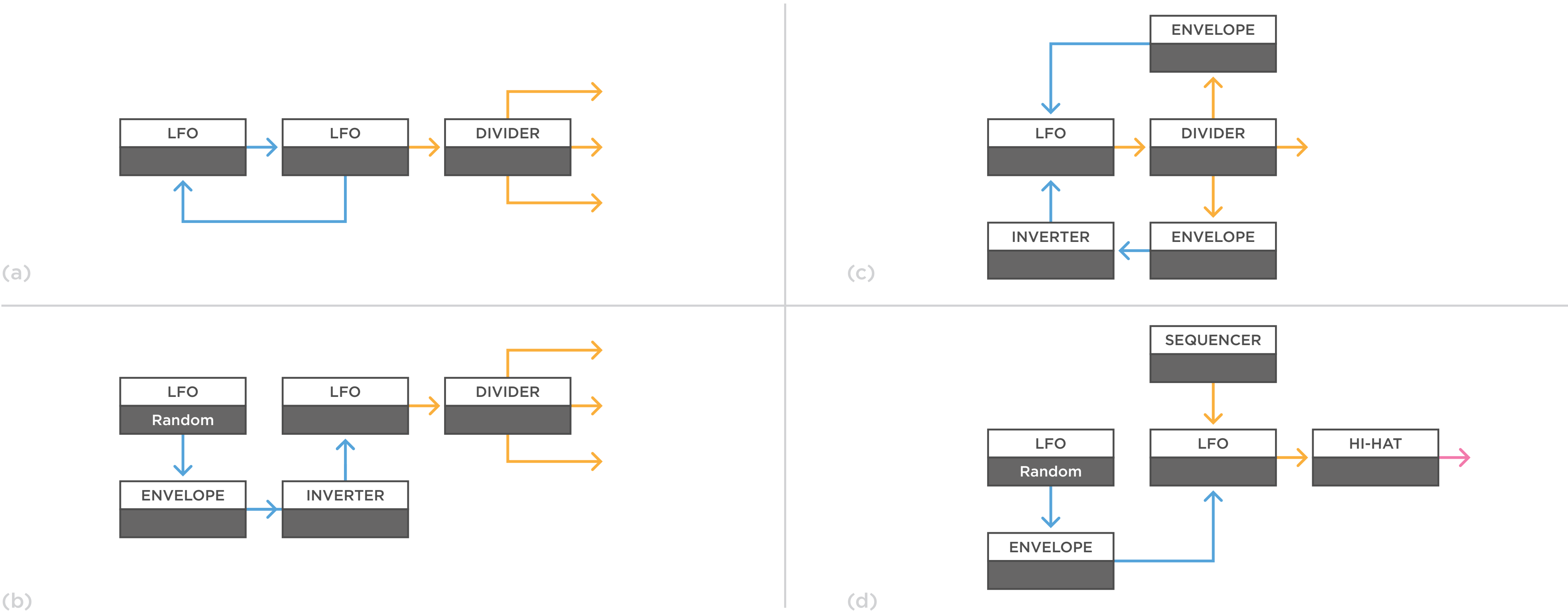
a - Let’s make our drone patch a little more complex. We introduce a simple envelope to control the filter and VCA. And we use a sequencer to trigger that envelope and send the oscillators a 1v/oct signal to create a sequence with more distinct notes. Then we use one LFO on the filter again, and the other to a VCA, controlling the amount of frequency modulation between oscillator one and two. This is a simple way to keep a simple sequence more dynamic.

b - Instead of using LFOs to create slow dynamic changes, we can also use them at audio rates to create subtle textures in sounds. For example, we can use one LFO at audio rates to subtly influence the filter, but instead of it running steady we can use a slowly changing smooth random voltage to have the LFO drift a bit over time. Or, instead of a random source, we can use a CV sequencer to program a sequence to set the speed of the LFO, creating rhythmic textures.

Voltage Controlled LFO

Dynamic Clock

Audio
Control Voltage
Trigger / Gate



a - You can use the square output of a LFO as a clock source. Send it to a clock divider, and you set up a steady clock with multiple divisions in your patch. If you introduce a second LFO and some cross-modulation between the two, you can create more chaos in the clock source.

b - You can also just use a single LFO and send it a signal every now and then to make little jumps, or temporary speed it up or down. For example, use a random output from a LFO to trigger a simple envelope. If you invert that signal before going to the LFO, the clock slows down sometimes.

c - You can also use clock division created by the clock itself to trigger envelopes. For example, use two different divisions, each triggering their own envelope, both influencing the core clock speed. One with a positive effect, the other with an inverted effect.

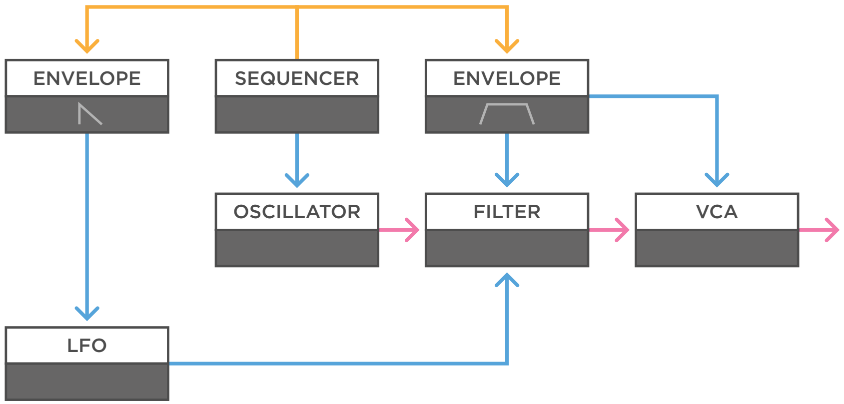
d - In this example we use a sequencer to retrigger a steady square LFO, that in turn triggers a hi-hat module. If we use a randomly triggered envelope to have the LFO speed up every now and then, we create random speeded up bursts of triggers in between a steady pattern.

Voltage Controlled LFO

Dynamic Voice

MONOTRAIL
TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)

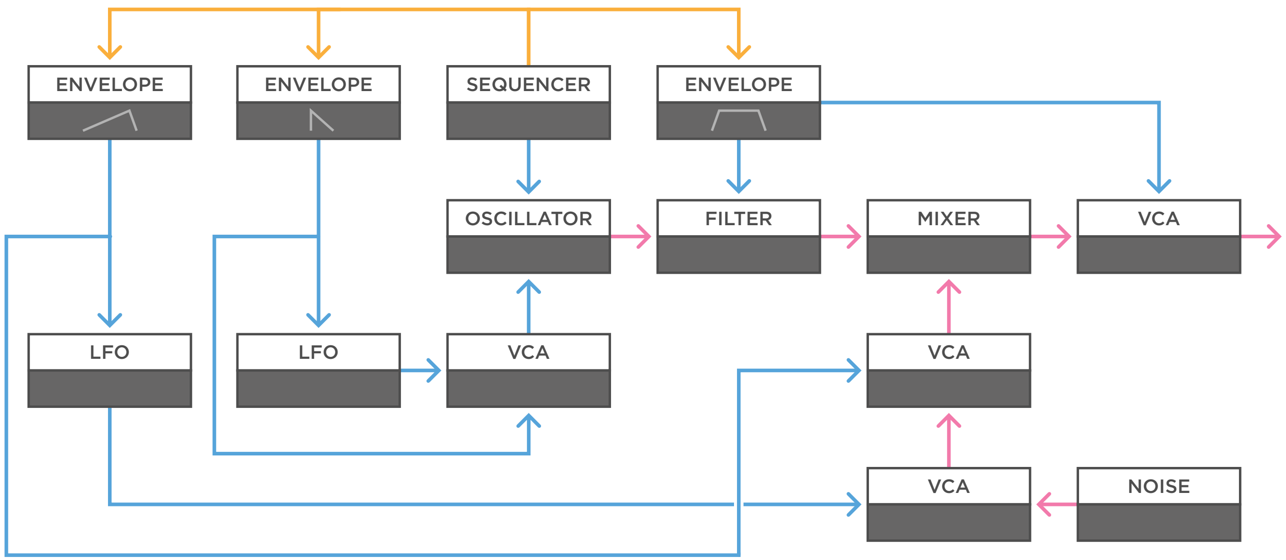
a - Let's use some modulation to create more interesting single melodic tones. Here we have one oscillator through a filter and a VCA. We use a sequencer to send the oscillator a 1v/oct signal, and to gate an attack hold release envelope going to both the filter and VCA, so we can play our voice and create longer sounds.

We add one LFO to influence the filter, but instead of it running steady, we multiply the sequencers gate output to another envelope with a short attack and slightly longer decay time. We send that envelope to influence the LFO speed. Now we can set our LFO to a speed that we like, but every time we start playing a note, it starts off a lot faster, and then quickly slows down, creating an interesting transient to the sound. Especially if we have it speed up the LFO to audio rates.

Voltage Controlled LFO

Dynamic Voice

Audio
Control Voltage
Trigger / Gate



(a)

a - You can expand on this simple principle and make it as complex as you like. For example, we can change our setup a bit by sending the LFO to the input of a VCA before going to the oscillators linear FM input. Then we multiply the envelope going to the LFO and send it to open that VCA. This way the envelope speeds up the LFO, as well as determines the amount of FM the LFO creates on the oscillator. Starting with a lot, and then quickly slowing down and fading out.

To incorporate another idea, let's trigger a third envelope with another copy of our gate signal. This time with a slow rising attack and some release. And send that envelope to another LFO, again having it speed up. We then send the LFO to open a VCA that we feed some white noise as audio. The result goes to another VCA, which in turn is opened by the same rising envelope. Finally, the noise is mixed with our synth voice. This way, if we hold a note for a longer time, we introduce a slowly rising amount of noise, created by a slowly faster going LFO.